

15. Centripetal force is responsible to

- (a) independent motion of the object in space
- (b) keep the body moving along the circular path
- (c) fly the object along a straight line
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (b)

Centripetal force is responsible for keeping an object moving in a circular path. When an object moves in a circular motion, it experiences an inward force directed towards the centre of the circle. This force is called the centripetal force.

$$\text{Centripetal force} = \frac{(\text{mass of the object}) \times (\text{velocity})^2}{\text{radius of the circular path}}$$

16. In the night of strong wind, dew is not formed because–

- (a) Rate of evaporation is fast
- (b) Moisture in air is low
- (c) Temperature is high
- (d) Sky is not clear

40th B.P.S.C. (Pre) 1995

Ans. (a)

Dew does not form in the night of strong wind because in the strong wind the rate of evaporation is fast. Thus, the particles of the dew tend to evaporate.

17. A fan produces a feeling of comfort during the hot weather because–

- (a) Fan supplies cool air.
- (b) Our body radiates more heat in air.
- (c) Conductivity of air increases.
- (d) Our perspiration evaporates rapidly
- (e) None of the above / more than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (d)

In the summer season, we feel comfortable due to rapid evaporation of perspiration coming out of the body by the fan.

18. If the door of refrigerator is left open for few hours, then the room temperature will

- (a) decrease
- (b) increase
- (c) remain the same

(d) decrease only in the vicinity of refrigerator

(e) None of the above / More than one of the above

67th B.P.S.C. (Pre) 2021

Ans. (b)

If the door of refrigerator is left open for few hours, then the room temperature will increase and room will become hot because refrigerator exhausts more heat into the room than earlier. Thus, a room cannot be cooled by keeping the door of refrigerator open.

19. What is condensation?

- (a) The change of gas into solid
- (b) The change of solid into liquid
- (c) The change of vapour into liquid
- (d) The change of heat energy into cooling energy
- (e) None of the above/More than one of the above

B.P.S.C. CDPO 2018

Ans. (c)

Condensation is the process where water vapour becomes liquid. It is the reverse action of evaporation, where liquid water becomes a vapour.

20. Cooling is done by –

- (a) Flow of water
- (b) Release of compressed gas
- (c) Cooking gas
- (d) Melting the solid

47th B.P.S.C. (Pre) 2005

Ans. (b)

Most of the air conditioners use compressed gas, which can cool the room or other places.

Wave Motion

1. An example of longitudinal wave is :

- (a) Radio wave
- (b) Sound wave
- (c) X-ray
- (d) Gamma ray
- (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (b)

Sound waves are characterized by the motion of particles in the medium and are longitudinal (in air and any fluid medium) mechanical waves while Radio-waves, X-rays and light waves are electromagnetic waves.

2. Sound wave in air is –

- (a) Transverse
- (b) Longitudinal